

LETTER TO THE JOURNAL

Vaginal delivery provides skin colonization resistance from environmental microbes in the NICU

Dear Editor,

Here, we present novel data suggesting there is an increased sharing of the environmental microbiome on the skin in infants delivered by cesarean section (CS) compared to vaginal delivery (VD) in the neonatal intensive care unit (NICU).

It has previously been shown that delivery mode is a major factor influencing early life skin bacterial colonization.^{1–3} Furthermore, skin microbiome differences by delivery mode correlate with variations in later skin barrier function and possibly health outcomes.² Although reports vary, CS delivery has been associated with an increased risk of developing atopic dermatitis compared to VD in a recent large meta-analysis.⁴ Additionally, variations in skin microbiome between infants delivered at different hospitals have been theorized to be due to variances in the microbes of the hospital environment.² Building on this, we hypothesized that the overall low skin microbial load in CS infants⁵ would provide less resistance to acquisition and colonization by environmental bacteria when compared to VD infants.

To address this, shotgun metagenomic sequencing was performed on skin swabs collected from behind the ear of 73 infants (16 delivered vaginally, 57 by CS) housed in the NICU during the first week of life as previously described.⁶ Environmental swabs from different areas of the NICU were also collected. Infants who were in the NICU were selected as they were postulated to potentially acquire more environmental bacteria due to a prolonged stay in the hospital and limited skin-to-skin contact with family members,⁷ making them the ideal cohort to preliminarily investigate this hypothesis. Differences in the skin microbiome between delivery mode and the environment were compared. SourceTracker v0.9.5 was applied to estimate the proportion of environmental bacterial taxa in the skin microbiome.⁸ See [Supplemental Data](#) for detailed methods.

CS infants were more likely to receive peripartum antibiotics ($p = .002$) and be female (.021) than VD infants (Table

S1). There were no differences in gestational age, infant antibiotic use, race or ethnicity between delivery modes.

Lower alpha diversity was observed in VD compared to CS (Shannon diversity index, $p = .002$, Figure 1A) with a clear difference in microbial composition between delivery modes (PERMANOVA, $p = .001$, Figure 1B). There were 86 significant differentially abundant bacterial genera between delivery modes. Notably, *Escherichia* and *Lactobacillus* were higher in VD, contrasted with increased *Pseudomonas*, *Staphylococcus* and *Acidovorax* in CS (Figure 1C). *Staphylococcus* and *Acidovorax* were the most distinguishing genera between delivery modes (Figure 1D). Level 4 KEGG analysis revealed lower Shannon diversity in VD (Figure 1E) with several differentially abundant pathways identified between delivery modes (Figure 1F), many involving lipid and antibiotic metabolism. Of potential interest, there was upregulation of fatty acid synthase in vaginal deliveries, with short chain fatty acids thought to have a role in the immune tolerance of the skin.⁹

The microbiome of the NICU environment clustered with the skin bacteria in those delivered by CS ($p = .001$, Figure 2A). SourceTracker analysis demonstrated a significantly greater sharing of environmental microbes in CS compared with VD infants ($p = .006$). These organisms predominantly were sourced from a nurses' station wash sink ($p = .007$, Figure 2B). The predominant taxa shared between the skin and wash sink belonged to the genus *Pseudomonas* (Figure 2C). There was also sharing of *Staphylococcus aureus* and species belonging to the genera *Corynebacterium* and *Cutibacterium*, each of which has previously been linked to psoriasis acne, and atopic dermatitis.¹⁰

In conclusion, there was increased sharing of the environmental microbiome on the skin of babies born by CS versus VD. We speculate that CS infants exhibited higher sharing of the environmental microbiome due to being closer to 'sterile' when delivered (as opposed to VD infants who are initially colonized predominantly by maternal

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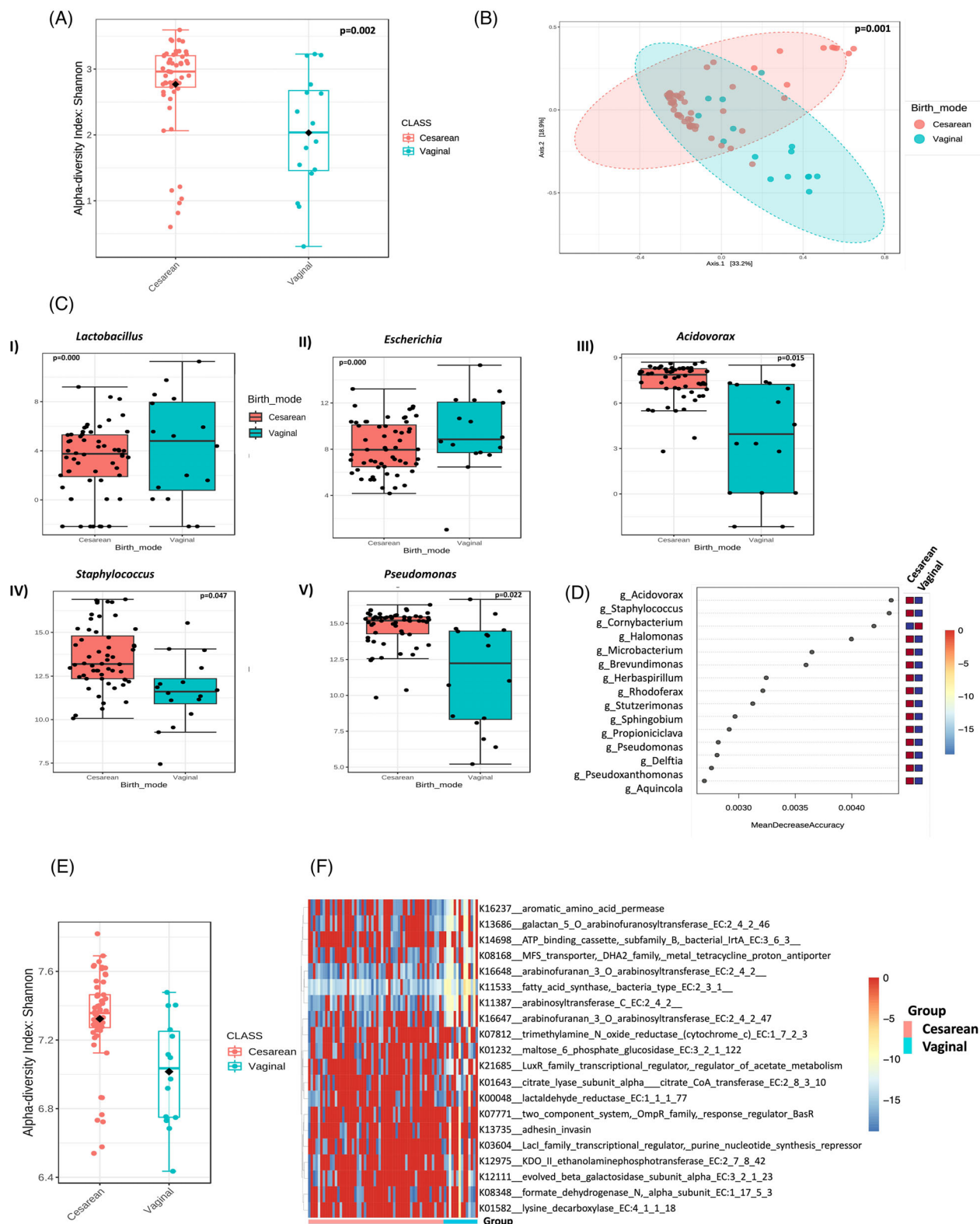


FIGURE 1 Skin microbiome differences between cesarean section (CS) and vaginal delivery (VD) infants. (A) Alpha diversity (Shannon index) of the skin microbiome between CS and VD infants. (B) principal coordinate analysis plot of the Bray-Curtis dissimilarity distances between the microbiomes of the CS and VD infants. (C) Bar plots showing differences between CS and VD infants for the genera: *Lactobacillus*, *Escherichia*, *Acidovorax*, *Staphylococcus* and *Pseudomonas*. (D) Random forest plot showing the most distinguishing taxa at the genus level between delivery modes. (E) Differences in alpha diversity of functional pathways (level 4 KEGG) between CS and VD. (F) Heatmap plot showing the top 20 functional enzymes based on their log₂ fold change values (level 4 KEGG).

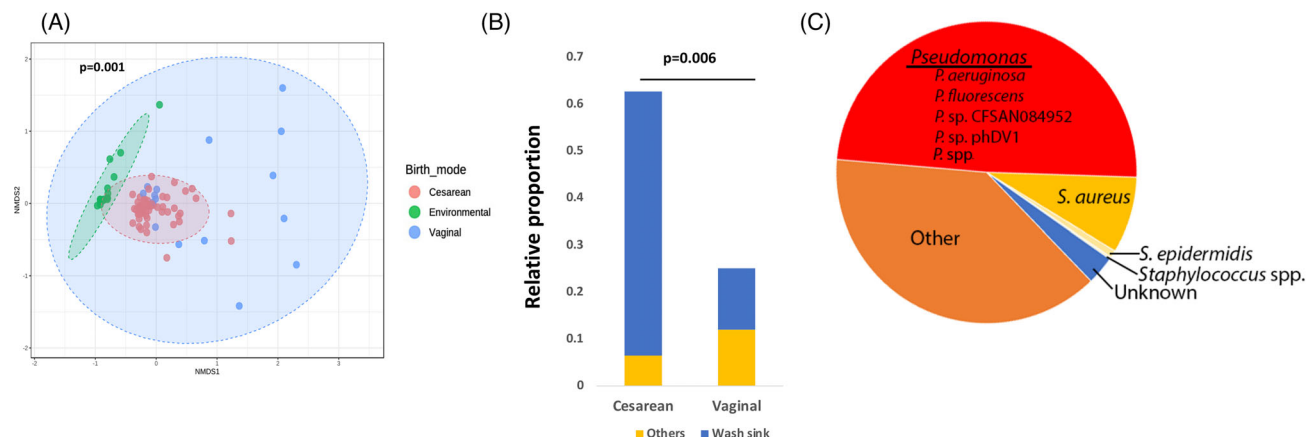


FIGURE 2 Sharing of the environmental microbiome by delivery mode. (A) Non-metric multidimensional scaling plot visualizing the Bray-Curtis dissimilarity matrix of the skin microbiome by delivery mode compared with environmental samples. (B) The relative proportion of environmental microbiome sources with the skin microbiome of subjects in those born by cesarean section (CS) versus vaginal delivery (VD). (C) Pie chart showing the most common taxa shared between the environment and infants' skin.

vaginal and fecal microbes,¹ as evidenced by higher levels of *Escherichia* and *Lactobacillus* in our study), thus having less colonization resistance and allowing opportunistic colonization by the environment. It is also possible that the higher levels of perinatal antibiotics in the CS group might contribute to the establishment of the pioneer species, by potentially reducing skin bacterial load further and allowing for more opportunistic colonization. It is difficult to separate this from the effects of CS, as perinatal antibiotics are received in all CS deliveries. Future studies looking at colonization between vaginal deliveries with and without perinatal antibiotics may help gain further insight into this. It is acknowledged that these results may not be generalizable to healthy full-term infants who are not admitted to the NICU, who have closer skin-to-skin contact with family members and different environmental exposures; this warrants further assessment. In addition, the overall sample size is limited especially for vaginal deliveries, and further investigation should include a larger cohort. Moreover, examining the microbiome of other potential sources of colonization such as the caretakers' microbiome and breast milk microbiome may help to tease apart other contributors. Lastly, the differences in functional pathways found between delivery modes are of potential interest with exploration in larger studies needed to see if this may have a lasting impact.

These preliminary data underscore the importance of the environment in the development of the skin microbiome. Most importantly, the impact of our findings on later health outcomes, including skin diseases associated with delivery mode, warrants further investigation.

AUTHOR CONTRIBUTIONS

Concept and design: Suchitra K. Hourigan and Ian A. Myles. **Acquisition, analysis or interpretation of data:** All

authors. **Drafting of the manuscript:** Suchitra K. Hourigan. **Critical revision of the manuscript for important intellectual content:** All authors. **Statistical analysis:** Prem Prashant Chaudhary, Shareef M. Dabdoub and Purnima S. Kumar. **Obtained funding:** Suchitra K. Hourigan and Ian A. Myles. **Supervision:** Suchitra K. Hourigan and Ian A. Myles. **Reviewed and approved the manuscript:** All authors.

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CONFLICT OF INTEREST STATEMENT

The authors have declared that no conflict of interest exists.

DATA AVAILABILITY STATEMENT

The data sets generated and/or analyzed in the current study are available in the NCBI SRA repository under Bio-Projects ID PRJNA854811 (infant data) and PRJNA417283 (environmental data).

ETHICS STATEMENT

The study was Institutional Review Board approved (WCG IRB 1300205), and written informed consent was obtained.

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REFERENCES

1. Dominguez-Bello MG, Costello EK, Contreras M, et al. Delivery mode shapes the acquisition and structure of the initial microbiota across multiple body habitats in newborns. *Proc Natl Acad Sci U S A*. 2010;107(26):11971-11975.
2. Rapin A, Rehbinder EM, Macowan M, et al. The skin microbiome in the first year of life and its association with atopic dermatitis. *Allergy*. 2023;78(7):1949-1963.
3. Bogaert D, van Beveren G, de Koff EM, et al. Mother-infant microbiota transmission and infant microbiota development across multiple body sites. *Cell Host Microbe*. 2023;31(3):447-460.e6.
4. Xiong Z, Zhou L, Chen Y, et al. Prevalence of eczema between cesarean-born and vaginal-born infants within 1 year of age: a systematic review and meta-analysis. *Eur J Pediatr*. 2022;181(6):2237-2247.
5. Mueller NT, Differding MK, Sun H, et al. Maternal bacterial engraftment in multiple body sites of cesarean section born neonates after vaginal seeding-a randomized controlled trial. *mBio*. 2023;14(3):e0049123.
6. Chaudhary PP, Myles IA, Zeldin J, et al. Shotgun metagenomic sequencing on skin microbiome indicates dysbiosis exists prior to the onset of atopic dermatitis. *Allergy*. 2023;78(10):2724-2731.
7. Hourigan SK, Dominguez-Bello MG. Microbial seeding in early life. *Cell Host Microbe*. 2023;31(3):331-333.
8. Knights D, Kuczynski J, Charlson ES, et al. Bayesian community-wide culture-independent microbial source tracking. *Nat Methods*. 2011;8(9):761-763.
9. Sanford JA, Zhang LJ, Williams MR, Gangoiti JA, Huang CM, Gallo RL. Inhibition of HDAC8 and HDAC9 by microbial short-chain fatty acids breaks immune tolerance of the epidermis to TLR ligands. *Sci Immunol*. 2016;1(4):eaah4609.
10. Sinha S, Lin G, Ferenczi K. The skin microbiome and the gut-skin axis. *Clin Dermatol*. 2021;39(5):829-839.

SUPPORTING INFORMATION

Additional supporting information can be found online in the Supporting Information section at the end of this article.